Perceptra: AI System for Analysing Public Human Behaviour

A Major Project/Internship Report Submitted for the requirement of

Final Year Semester-VIII

Electronics and Telecommunication Engineering

by

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Certificate

This is to certify that, the Major Project/Internship titled

"Perceptra: AI System for Analysing Public Human Behaviour"

is a bonafide work done by

SHIVAM SINGH (Roll No 21ET1014)

and is approved for the practical work done during Semester - VIII in

Major Project/Internship
in
Electronics and Telecommunication Engineering
to the
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Supervisor	Project Coordinator
Head of Department	Principal

Certificate of Approval by Examiners

This is to certify that the submission entitled for the project "**Perceptra: AI System for Analysing Public Human Behaviour**" is a bonafide work done by **SHIVAM SINGH** under the guidance of **Dr. VIJAY DAHAKE**. This Project is approved in the partial fulfillment of the requirement for degree of ***B.tech in Electronics and Telecommunication Engineering*** Semester-VIII, D. Y. Patil Deemed to be University.

Internal Examiner :

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DECLARATION

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Abstract

This project presents a prototype of an AI-based surveillance system for real-time detection of unattended objects such as bags at Mumbai local train stations. By using deep learning algorithms and computer vision, the system continuously monitors CCTV footage to identify when a person leaves behind a suspicious item and moves away.

The model integrates YOLOv8 (You Only Look Once Version 8) for fast and efficient object detection. The logic differentiates between attended and unattended objects using bounding box proximity and time thresholds. If a bag is left unattended for more than a few seconds, an alert is generated, and a laser pointer can mark the object for security attention.

Additional features include:

-Real-time video feed processing
-Behavior tracking to cancel alerts when the owner returns
-Future scope for stalking detection and women's safety features.

The system can be deployed as a standalone AI device connected to existing DVR setups, helping to improve public safety through intelligent monitoring

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Chapter 1

Introduction

In recent years, ensuring public safety at crowded places like railway stations has become a major concern. The rise in threats, unattended baggage, and suspicious activities requires constant human monitoring, which is often inefficient and error-prone.

This project aims to build an AI-powered system capable of real-time monitoring of CCTV feeds to detect unattended objects and issue alerts. By integrating computer vision and machine learning, we propose a solution that automates surveillance and reduces human error in threat detection.

1.1 Problem Definition

The Mumbai local train network is among the busiest in the world, with lakhs of passengers traveling daily. Due to high footfall and limited surveillance personnel, identifying potential security threats like unattended bags becomes challenging.

There is a need for an automated, intelligent surveillance system that can:

Detect objects left unattended for a set duration.

Identify and associate persons with objects.

Generate real-time alerts for security teams.

Assist in improving passenger safety without manual monitoring.

1.2 Organization of Report

his project report is organized into the following chapters:

Chapter 1: Introduction This chapter provides an overview of the project, including background, motivation, objectives, and scope of the system developed for surveillance in Mumbai local stations.

Chapter 2: Literature Survey This section discusses existing systems and technologies used in surveillance and object detection, along with a review of YOLO models and similar AI-based systems.

Chapter 3: System Analysis and Requirements This chapter outlines the system requirements, both hardware and software, along with system specifications and problem definition.

Chapter 4: System Design and Methodology This section covers the architecture of the system, flowcharts, data flow diagrams, and detailed design of the AI model and how it interacts with camera inputs and the alert system.

Chapter 5: Implementation Details of implementation such as the AI model integration, Python scripts, YOLOv8 usage, webcam and RTSP integration, sound alerts, and bounding box detection are presented here.

Chapter 6: Results and Discussion This chapter presents the outputs of the system — including screenshots, images of object detection in real-time, system behavior, and limitations observed.

Chapter 7: Conclusion and Future Scope This section summarizes the project and discusses potential improvements, including women's safety, threat detection, and advanced pose-based behavior analysis.

References A list of research papers, documentation, and websites referred to during the development of the project.

Appendix Includes sample code, system setup instructions, and other supporting material relevant to the project.

Chapter 2

Literature Survey

Several studies have explored AI in surveillance, including behavior analysis, object detection, and public safety enhancements:

YOLO (You Only Look Once) is a state-of-the-art object detection model used widely in surveillance.

OpenCV provides real-time computer vision capabilities and video feed handling.

Pose estimation models like OpenPose can assist in detecting body posture for behavior analysis.

Some existing systems detect motion or rely on basic object tracking but lack intelligent behavior analysis or unattended object logic.

This project improves on previous work by combining object detection and tracking logic for real-time unattended object alerts. [1].

Chapter 3

System Design

3.1 System Architecture

The system is designed to perform real-time surveillance using AI algorithms. The main components are:

- Input: CCTV video feed (via webcam or RTSP stream).
- Object Detection Module: YOLOv8 model to detect people and baggage (handbags, backpacks, suitcases).
- Tracking Module: Tracks the detected bags and their associated persons.
- Logic Layer: Determines if a bag has been left unattended for a specific duration (e.g., 10 seconds).
- Alert System: If a bag is unattended, an alert is generated (with buzzer sound and bounding box).
- User Interface (UI): Displays annotated video with grid layout for multiple feeds.

3.2 Flow Chart

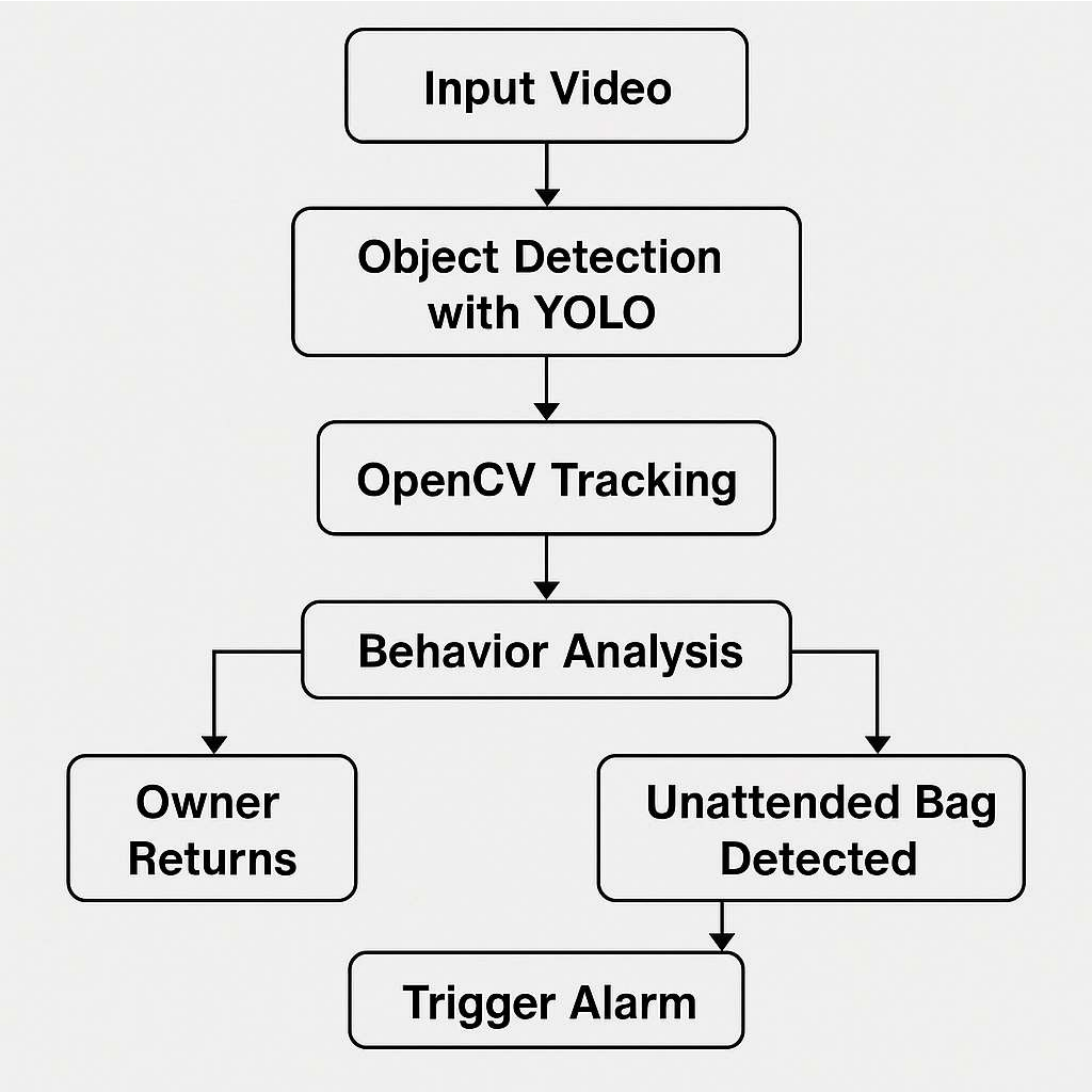

Figure 3.1: Flowchart

Chapter 4

Hardware and Software Requirements

Hardware

Category	Requirements
Hardware Requirements	
Processor	Minimum Intel i5 / AMD Ryzen 5 or equivalent
RAM	8 GB (16 GB recommended)
Storage	100 GB Free Disk Space (SSD preferred)
GPU	NVIDIA GPU with CUDA support (e.g., GTX 1650 / RTX 2060)
Camera	CCTV / USB Web Camera / RTSP IP Camera
Laser Pointer (optional)	Red laser pointer for visual object marking
Speaker / Buzzer	For alert sound notifications

Table 4.1: Hardware

Software

Category	Requirements
Software Requirements	
Operating System	Windows 10 / 11 or Linux (Ubuntu 20.04 or above)
Programming Language	Python 3.8 or higher
Libraries	OpenCV, Ultralytics YOLOv8, NumPy, Playsound, Matplotlib
Development Environment	Visual Studio Code / PyCharm / Jupyter Notebook
Other Tools	CUDA Toolkit (if using GPU), pip, Git

Table 4.2: Software

Chapter 5

Implementation

The project was implemented using Python and Ultralytics YOLOv8.

5.1 Object Detection and Tracking

- The YOLOv8 model was used to detect specific classes: person, backpack, handbag, suitcase.
- Detected objects are tracked across frames using object IDs.
- A logic was applied to check if any detected bag has no associated person nearby.

5.2 Alert Mechanism

- If a bag remains unassociated for more than 10 seconds, it is marked as unattended.
- audio alert is triggered and a red bounding box is shown on the screen.
- If the owner returns and picks up the bag, the alert is cancelled.

5.3 Multi-Camera Grid Layout

- RTSP links or multiple webcams can be shown in a 2×2 grid.
- Each feed is processed sequentially to minimize resource usage.
- Alerts are logged with timestamps.

Chapter 6

Results and Conclusion

System Output

The AI-based surveillance system was successfully implemented and tested on live camera feed and pre-recorded video samples. The system was able to detect the following:

- Person and bag detection using YOLOv8.
- Identification of unattended bags using bounding box disassociation.
- Real-time alert generation when a bag was left unattended for more than the defined threshold.
- Sound alerts triggered for security personnel.
- Cancelation of alerts when the owner returned to retrieve the bag.

Screenshots and Output Visualization

Figure 6.1 shows the detection of a person and a bag in real-time using the CCTV feed.

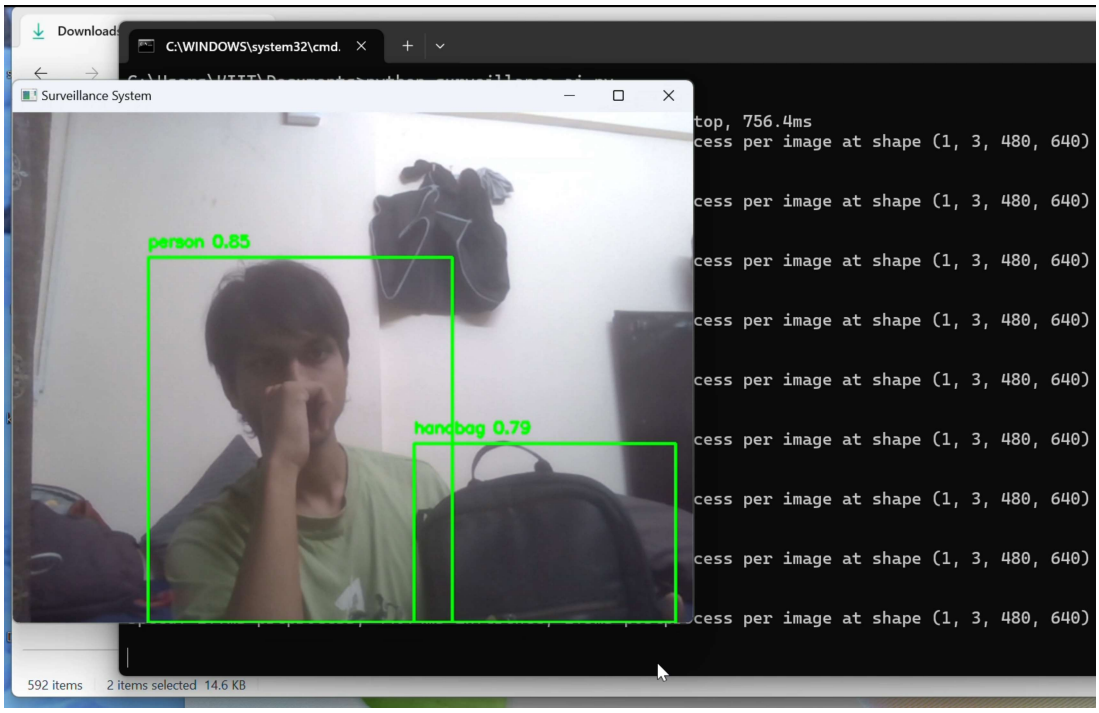

Figure 6.1: Person and Bag Detection using YOLOv8

Figure 6.2 illustrates the case where the person moves away from the bag and an alert is triggered, including a bounding box and alert message on screen.

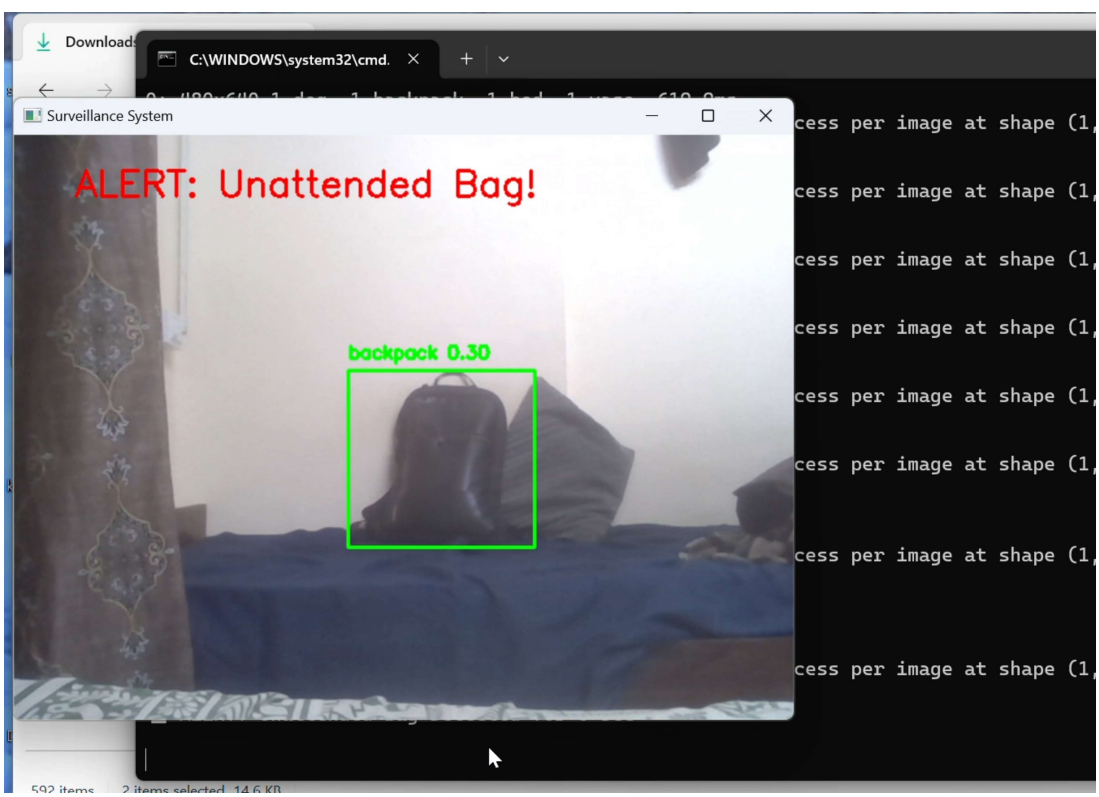

Figure 6.2: Alert Triggered for Unattended Bag

Figure ?? demonstrates the buzzer alert and display warning when an unattended

object is detected for more than the allowed time.

Performance Discussion

- The object detection model (YOLOv8) performed with high accuracy for person and bag categories.
- False positives were initially encountered when bag and person bounding boxes slightly separated — this was improved by checking for complete separation.
- Latency was minimal on GPU systems and acceptable on CPU for a pilot prototype.
- Sequential processing supported up to 4 CCTV feeds with a delay of 4–5 seconds between each frame processing.

Limitations

- Model requires a high-performance GPU for smooth parallel feed handling.
- Detection of owner return is based on visual proximity and may fail if occlusion occurs.
- System cannot distinguish between a threat and a harmless object — further classification models are needed.

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Table 6.1: Details of Student

Future Scope

- **Advanced Behavior Recognition:** Incorporate human pose estimation techniques such as OpenPose to detect activities like stalking, fighting, or falling, enabling more contextual alerts.
- **Women's Safety Enhancements:** Add focused AI modules to monitor safety in and around ladies' compartments or platforms, including detection of prolonged loitering or unauthorized male proximity.
- **Owner Re-identification:** Integrate facial recognition or object re-identification algorithms to cancel alerts if the original bag owner returns, minimizing false positives.
- **Multicamera Real-Time Processing:** Optimize the system for real-time parallel processing of multiple CCTV feeds using edge computing devices or high-performance GPUs.

- **Custom Object Detection Models:** Train and fine-tune custom models for local scenarios like suspicious objects, restricted area intrusion, or illegal activities.
- **Mobile Alerts and Cloud Integration:** Push notifications to mobile apps or security dashboards for faster response, and store logs securely in the cloud for auditing and analytics.
- **Government-Scale Deployment:** After successful pilot testing, propose integration with Indian Railways, Smart City projects, and metro security systems.
- **Standalone Hardware Module:** Develop a dedicated AI-enabled standalone device that can be attached to existing DVR/NVR systems at stations without requiring infrastructure overhaul.

Bibliography

[1] OpenCV Documentation:. https://docs.opencv.org.

[2] Ultralytics YOLOv8 Documentation: https://docs.ultralytics.com

[3] PyTorch: https://pytorch.org

Acknowledgments

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Date: _____